Gamma Function for Non-Integer Factorials

1 Factorial recurrence

For a natural number n, $n! = 1 \cdot 2 \cdot 3 \cdot ... \cdot n$. We also define $0! = 1$, and with this convention factorials satisfy a distinct recurrence property of the next factorial obtained from the previous by multiplying by the next integer: $(n+1)! = (n+1) \cdot n!$

A natural question is: how could factorial be extended to quantities like $\left(\frac{1}{2}\right)!$ or 1.7!? The problem of extending factorials was studied by Daniel Bernoulli and Leonhard Euler. Adrien-Marie Legendre introduced the modern symbol $\Gamma(x)$ for the standard extension, the Gamma function. This notation embeds a shift:

$$\Gamma(n+1) = n! \text{ for } n = 1, 2, 3, \ldots \text{ so more precisely, } n! = \Gamma(n+1).$$

This extension of factorials represents a smooth function that satisfies recurrence $\Gamma(x+1) = x\Gamma(x)$, or equivalently, using the exclamation symbol, $x! = x \cdot (x-1)!$.

2 The Gamma Function

Definition

$$\text{For } x > 0,\ \Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt.$$

This integral represents an area under a curve. If this area is finite, then the quantity is a real number. If the area is infinite, then the formula would not define an ordinary finite value.

That is why the convergence of the integral is important. Before using the function defined with the help of the integral, we will check that the area it describes is finite. In two ways, this is an improper integral. First, because the interval begins at $t = 0$ where t^{x-1} is not defined for *every* $x > 0$, and second, because is extends to infinity.

Convergence of the integral

First, we will prove that the integral defining $\Gamma(x)$ converges for every $x > 0$:

$$\int_0^\infty t^{x-1} e^{-t} dt = \int_0^1 t^{x-1} e^{-t} dt + \int_1^\infty t^{x-1} e^{-t} dt.$$

For $t \in (0, 1]$, $e^{-t} \leq 1$, and $0 \leq t^{x-1} e^{-t} \leq t^{x-1}$ holds for the integrand. Since $x > 0$, integral $\int_0^1 t^{x-1} dt = \frac{t^x}{x}\Big|_0^1 = \frac{1}{x}$, which is finite for any given $x > 0$. Thus, the integral will converge near 0.

For the tail integral, we will use the fact that the exponential decay dominates polynomial growth as t approaches infinity, for which reason $t^{x-1} e^{-\frac{t}{2}}$ is bounded on $[1, \infty)$. Denoting the bound C we have for $t \geq 1$:

$$0 \leq t^{x-1} e^{-t} = (t^{x-1} e^{-\frac{t}{2}}) \cdot e^{-\frac{t}{2}} \leq C \cdot e^{-\frac{t}{2}}.$$

Since integral $\int_1^\infty e^{-t/2} dt$ converges, the Gamma function integral also converges on $[1, \infty)$, and the Gamma function is defined; see Figure 1 for its graph and the value of $\Gamma(4)$. Notice that the graph passes through factorial values shifted by one unit: $\Gamma(1) = 0!$, $\Gamma(2) = 1!$, $\Gamma(3) = 2!$, and so on.

Scaling identity



Figure 1: Desmos graph of the Gamma function with marked point (4,6). Values for negative x are shown for illustration only.

For the Beta function, the following equality is important.

$$\text{If } x > 0 \text{ and } s > 0, \text{ then } \int_0^\infty t^{x-1}e^{-st}dt = \frac{\Gamma(x)}{s^x}. \text{ Equivalently, } \Gamma(x) = s^x \int_0^\infty t^{x-1}e^{-st}dt.$$

The one-line proof uses the substitution $u = st$. Then $t = \frac{u}{s}$ and $dt = \frac{du}{s}$, and since $s > 0$, the limits remain 0 and ∞: when $t = 0$, $u = 0$, and when t approaches infinity, u also approaches infinity. Therefore

$$\int_0^\infty t^{x-1}e^{-st}dt = \int_0^\infty \left(\frac{u}{s}\right)^{x-1} e^{-u}\frac{du}{s} = \frac{1}{s^x}\int_0^\infty u^{x-1}e^{-u}du = \frac{\Gamma(x)}{s^x}.$$

The functional equation

The Gamma function is important because it satisfies the relationship similar to factorial recurrence. Using integration by parts, we will prove the functional equation for all $x > 0$

$$\Gamma(x+1) = x\Gamma(x).$$

Let $u(t) = t^x, dv = e^{-t}dt$, and we get $du = xt^{x-1}dt$ with $v(t) = -e^{-t}$.

$$\Gamma(x+1) = \int_0^\infty t^x e^{-t}dt = \int_0^\infty udv = uv\Big|_0^\infty - \int_0^\infty vdu = \left(-t^x e^{-t}\right)\Big|_0^\infty + x\int_0^\infty t^{x-1}e^{-t}dt = 0 + x\Gamma(x).$$

The term $\left(-t^x e^{-t}\right)\Big|_0^\infty = 0$ since for $x > 0$, $\lim_{t\to 0+} t^x e^{-t} = 0$ as $0 < e^{-t} \leq 1$, and $\lim_{t\to\infty} t^x e^{-t} = 0$.

Extension of factorial

Using above functional equation, we can now prove that for all integers $n \geq 0$,

$$\Gamma(n+1) = n!$$

First, $\Gamma(1) = \int_0^\infty e^{-t}dt = 1 = 0!$ Then, by repeatedly using $\Gamma(x+1) = x\Gamma(x)$, we get

$$\begin{aligned}\Gamma(n+1) &= n\Gamma(n) = \\ &= n(n-1)\Gamma(n-1) = \\ &= \cdots = \\ &= n(n-1)\cdot ... \cdot 1 \cdot \Gamma(1) = \\ &= n!\end{aligned}$$

For example, $5! = \Gamma(6) = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot \Gamma(1) = 120$.

The factorial of one-half

To determine the value discussed in the beginning, we will use the functional equation, and first compute the value of $\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-\frac{1}{2}} e^{-t}dt = \sqrt{\pi}$. Using the substitution

$$t = u^2, dt = 2udu, \text{ and since } u > 0,\ t^{-\frac{1}{2}} = \frac{1}{u}.$$

since t goes from 0 to infinity, u also goes from 0 to infinity. Thus we have

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty \frac{1}{u} e^{-u^2} 2udu = 2\int_0^\infty e^{-u^2}du = \int_{-\infty}^\infty e^{-u^2}du.$$

This is the Gaussian integral, and we can use Fubini's theorem and polar coordinates with substitution $r^2 = x^2 + y^2, dA = rdrd\theta$:

$$\int_{-\infty}^\infty e^{-u^2}du = \sqrt{\left(\int_{-\infty}^\infty e^{-u^2}du\right)^2} = \sqrt{\left(\int_{-\infty}^\infty e^{-x^2}dx\right)\left(\int_{-\infty}^\infty e^{-y^2}dy\right)} = \sqrt{\int\int_{\mathbb{R}^2} e^{-(x^2+y^2)}dA} =$$

$$= \sqrt{\int_0^{2\pi}\int_0^\infty e^{-r^2} rdrd\theta} = \sqrt{\int_0^{2\pi} d\theta \int_0^\infty e^{-r^2} rdrd\theta}$$

The last substitution is $z = r^2$ with $\frac{1}{2}dz = rdr$:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\int_0^{2\pi} d\theta \int_0^\infty e^{-r^2} rdrd\theta} = \sqrt{2\pi \cdot \frac{1}{2}\int_0^\infty e^{-z}dz} = \sqrt{\pi \cdot 1} = \sqrt{\pi}.$$

Using the functional equation,

$$\left(\frac{1}{2}\right)! = \Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}.$$

By extension, $\left(-\frac{1}{2}\right)! = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, $\left(\frac{3}{2}\right)! = \Gamma\left(\frac{5}{2}\right) = \frac{3\sqrt{\pi}}{4}$, and $\left(\frac{5}{2}\right)! = \Gamma\left(\frac{7}{2}\right) = \frac{15\sqrt{\pi}}{8}$.

In general, half-integer factorials for $n = 0, 1, 2, ...,$ are

$$\boxed{\left(n - \frac{1}{2}\right)! = \Gamma\left(n + \frac{1}{2}\right) = \frac{(2n)!}{4^n \cdot n!}\sqrt{\pi}.}$$

3 Stirling's Formula for Factorials

Stirling's approximation

The asymptotic approximation for factorials named after James Stirling is

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

So, the ratio of both sides approaches one as n approaches infinity:

$$\lim_{n\to\infty} \frac{n!}{\sqrt{2\pi n}\cdot(n/e)^n} = 1.$$

This approximation is powerful because factorials grow extremely quickly, and Striling's formula provides a simpler expression close to $n!$ for sufficiently large n.

For the proof below, we will represent the formula as follows:
for each n $\exists$ ω_n with $|\omega_n| \leq 1$ such that $n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n}\left(1+\omega_n\sqrt{\frac{2}{\pi n}}\right)$.
This formulation provides the main term and also indicates that the relative error is at most of the order of $\frac{1}{\sqrt{n}}$.
To understand the motivation, we will first notice that for $n! = \Gamma(n+1)$, the integrand $t^n e^{-t}$ has a maximum at $t = n$, because

$$\frac{d}{dt}(t^n e^{-t}) = nt^{n-1}e^{-t} - t^n e^{-t} = (n-t)x^{n-1}e^{-t}, \text{ which has a zero at } t = n.$$

After factoring out the maximum,

$$t^n e^{-t} = \left(\frac{n}{e}\right)^n \left(\frac{t}{n}\right)^n e^{n-t}, \text{ and}$$

$$n! = \Gamma(n+1) = \left(\frac{n}{e}\right)^n \int_0^\infty \left(\frac{t}{n}\right)^n e^{n-t}dt.$$

We choose new variable s so that $\left(\frac{t}{n}\right)^n e^{n-t} = e^{-s^2}$, and taking natural logarithm of both sides gives

$$-s^2 = n\cdot ln\left(\frac{t}{n}\right) + n - t, \text{ or } s^2 = t - n - n\cdot ln\left(\frac{t}{n}\right).$$

Selecting $s < 0$ when $t < n$ and $s > 0$ when $t > n$, with this sign conversion, $t \in (0,\infty)$ corresponds to $s \in (-\infty,\infty)$.

Differentiating $-s^2 = n\cdot ln\left(\frac{t}{n}\right) + n - t$, we get $-2s\frac{ds}{dt} = \frac{n}{t} - 1 = \frac{n-t}{t}$, hence $\frac{dt}{ds} = 2s\frac{t}{t-n}$. Taylor's theorem for logarithm at 1 gives $ln(1+u) = u - \frac{u^2}{2(1+\theta u)^2}$ for some $\theta \in (0,1)$. With $u = \frac{t-n}{n}$, we have:

$$s^2 = t - n - n\cdot ln\left(\frac{t}{n}\right) = \frac{n}{2}\frac{u^2}{(1+\theta u)^2}, \text{ or } s = \frac{\sqrt{n/2}}{\frac{n}{t-n}+\theta}, \text{ and } \frac{t}{t-n} = \frac{n}{t-n} + 1 = \frac{\sqrt{n/2}}{s} + 1 - \theta.$$

Now we apply $\frac{t}{t-n} = \frac{\sqrt{n/2}}{s} + 1 - \theta$ to $\frac{dt}{ds} = 2s\frac{t}{t-n}$:
$\frac{dt}{ds} = 2s\frac{t}{t-n} = \sqrt{2n} + 2s(1-\theta)$ with θ depending on t and therefore s is between 0 and 1.
Returning to integration, we obtain

$$n! = \Gamma(n+1) = \left(\frac{n}{e}\right)^n \int_{-\infty}^{\infty} e^{-s^2}\left[\sqrt{2n} + 2s(1-\theta(s))\right]ds =$$

$$= \left(\frac{n}{e}\right)^n \left(\sqrt{2n}\int_{-\infty}^{\infty} e^{-s^2}ds + \int_{-\infty}^{\infty} e^{-s^2}2s(1-\theta(s))ds\right)$$

This equals to $\left(\frac{n}{e}\right)^n \left(\sqrt{2n\pi} + \int_{-\infty}^{\infty} e^{-s^2} 2s(1-\theta(s))ds\right)$. However, since $0 < \theta(s) < 1$, the absolute value of the remaining integral is less than $\int_{-\infty}^{\infty} e^{-s^2} 2|s|ds = 2$ which means that $\exists \omega_n : |\omega_n| \leq 1$ such that

$$n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n} \left(1 + \omega_n \sqrt{\frac{2}{\pi n}}\right).$$

Example

For example, using Stirling's approximation, $10! \approx \sqrt{20\pi} \left(\frac{10}{e}\right)^{10} \approx 3.60 \times 10^6$, while the exact value is $3,628,800$.

4 Applications to Geometry, Combinatorics, and Probability

Geometry

The volume of unit sphere in n-dimensional space is $V_n = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)}$.

The unit sphere is the set of points in n dimensions such that:

$$x_1^2 + x_2^2 + ... + x_n^2 \leq 1.$$

For $n = 2$, this is the unit circle area equaling $\frac{\pi^{2/2}}{\Gamma\left(\frac{2}{2}+1\right)} = \pi$.

For $n = 3$, unit sphere volume is $\frac{\pi^{3/2}}{\Gamma\left(\frac{3}{2}+1\right)} = \frac{\pi^{3/2}}{\frac{3\sqrt{\pi}}{4}} = \frac{4\pi}{3}$.

For $n = 4$, $V_4 = \frac{\pi^2}{\Gamma(3)} = \frac{\pi^2}{2}$.

Combinatorics

Striling's formula helps estimate $2n$ choose n for large n:

$$\boxed{\binom{2n}{n} = \frac{(2n)!}{(n!)^2} \sim \frac{4^n}{\sqrt{\pi n}}}.$$

This is because:

$$\frac{(2n)!}{(n!)^2} \sim \frac{\sqrt{4\pi n}\left(\frac{2n}{e}\right)^{2n}}{\left(\sqrt{2\pi n}\left(\frac{n}{e}\right)^n\right)^2} = \frac{4^n}{\sqrt{\pi n}}$$

Probability and Statistics: the Beta function

Beta distribution is an important distribution in statistics and probability theory. It is used, for example, to model online advertisement click-through rates. Beta function is used to normalize this distribution's density function so that area under the density function curve is exactly 1, since total probability cannot exceed 1.

For $\alpha > 0$ and $\beta > 0$ the function is defined as $B(\alpha, \beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx$.

Equivalently, $B(\alpha, \beta) = \int_0^\infty \frac{\tau^{\alpha-1}}{(1+\tau)^{\alpha+\beta}}d\tau$.

The most important connection between the Beta function and the Gamma function is the Beta-Gamma identity

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

This leads to the recurrence of the Beta function: $B(\alpha+1, \beta+1) = \frac{\alpha\beta}{(\alpha+\beta)(\alpha+\beta+1)}B(\alpha, \beta)$.

Example

For example, $B(3,4) = \frac{2\cdot 3}{5\cdot 6} \cdot B(2,3) = \frac{1}{5} \cdot \frac{\Gamma(2)\Gamma(3)}{\Gamma(5)} = \frac{1}{5} \cdot \frac{1!\cdot 2!}{4!} = \frac{1\cdot 2}{5\cdot 24} = \frac{1}{60}$, which first applies the recurrence and then the definition of the Beta-Gamma identity.

Bibliography

Rudin, Walter. *Principles of Mathematical Analysis.* 2nd ed. New York: McGraw-Hill, 1964.